General

Walk.mat contains the following channels:

- Ch1: time
- Ch2-161: ECoG 1-160
- Ch162: DI (Photodiode Feedback)
- Ch163: StimCode
- Ch164: GroupId

Note that the data were recorded at 1200 Hz and are not referenced to a particular electrode.

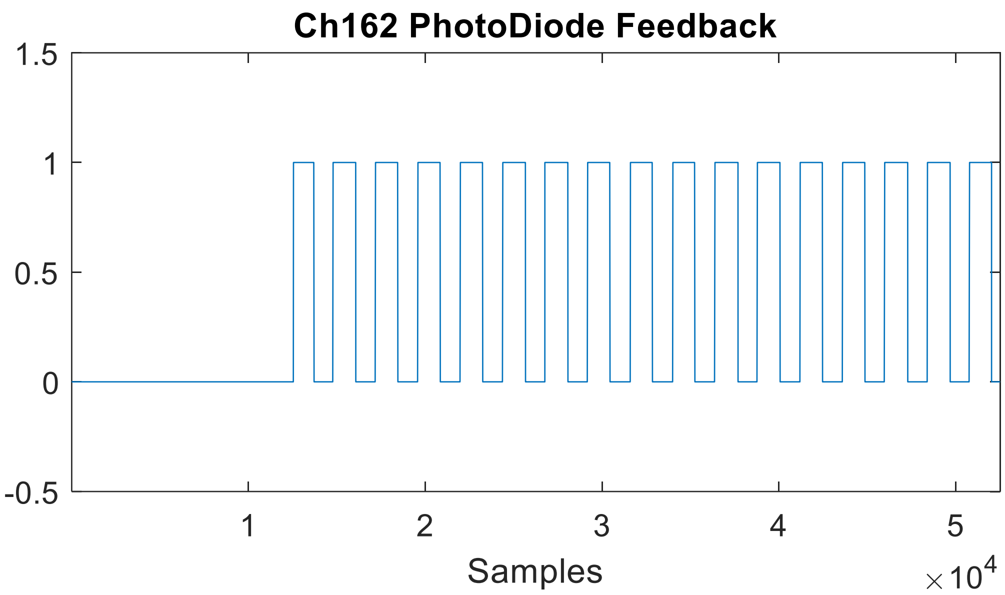
ECoG channels which go from Ch2 to Ch161 in the dataset, you can make sense of by checking the `Walk_channel_assignment.png` picture. Note that in the picture the channels go from 1 to 160.

Ch162 through Ch164 provide you with the information to link what happened in the Video to what is happening in the ECoG data. In the hospital setting, where this data was recorded, the patient looked at a computer screen which played the video (Walk.mp4).

Ch162 – DI (Photo Diode Feedback)

In order to have very precise information how the ECoG data aligns to the video, a photo diode was placed over the bottom left side of the screen, capturing the blinking square. The photo diode captures whenever this square changes from white to black and this is reflected in Ch162 (“1” = white square, “0” = black square).

Note that in the image below only a portion of the whole data is plotted for Ch162.

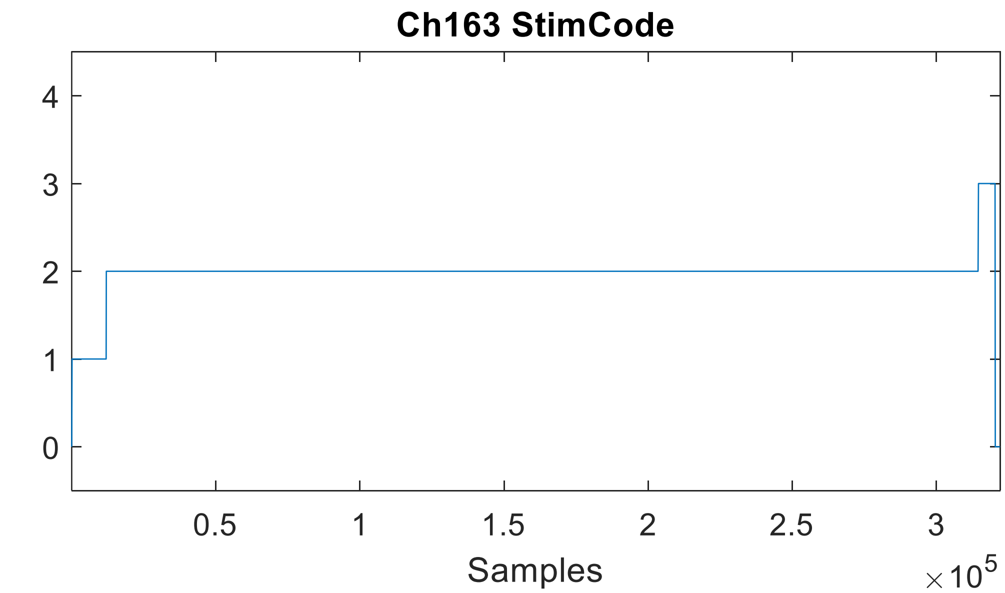


Ch163 – StimCode

In order to make sense of the `StimCode`, we need to check the data which is available in: `Walk_paradigmInfo.mat`. Specifically, this file contains a table called `taskInfo` with the following entries:

NumId	StringId	Label	Group
1	'ST_PreParadigm'	''	0
2	'ST_Video'	''	1
3	'ST_PostParadigm'	''	0

The `StimCode` channel reflects the `NumId` column in the `taskInfo` variable and indicates when a given part of the paradigm was played.



Ch164 – GroupId

In order to make sense of the `GroupId`, we need to check the data which is available in: `Walk_paradigmInfo.mat`. Specifically, this file contains a table called `taskInfo` with the following entries:

NumId	StringId	Label	Group
1	'ST_PreParadigm'	''	0
2	'ST_Video'	''	1
3	'ST_PostParadigm'	''	0

The `GroupId` channel reflects the `Group` column in the `taskInfo` variable and indicates when the video was played, as it is only 1 when the video was played and otherwise 0.